

Edam Hamza

✉ hamzaedam01@gmail.com — 📄 Edam Hamza — 🌐 EdamH — 🌐 edamhamza.vercel.app

Summary — ICT Engineer from SUP'COM (High Honors) and **GenAI Engineer at Converty**, building production **LLM systems**: multi-agent pipelines, RAG and vector search, generation pipelines, and MCP integrations, all shipping with billing, rate limiting, and security wired in from day one.

Education

Higher School of Communication of Tunis (SUP'COM) 2022 - 2025
ICT Engineering Degree, High Honors. Relevant Coursework: Big data, AI/ML, NLP, Cloud, Data Engineering

Preparatory Institute for Engineering Studies of Sfax 2020 - 2022
National Engineering School Entrance Exams : Math and Physics - Rank: 89/1831

Experience

Converty Oct 2025 – Present
AI Software Engineer, Platform & Development Automation
Technologies: LLMs, RAG, Vertex AI, Gemini, Qdrant, Vercel AI SDK, MCP, TypeScript, Node.js

- Designed and built **URANUS**, the in-house, provider-agnostic AI layer running **12 agents** across 4 families, so far, on one runner and hook pipeline; agents run on any provider through the Vercel AI SDK (Vertex AI and OpenAI live).
- Architected **3-layer rate limiting** (IP, session, per-store) and multi-tier caching in **Redis**, plus prompt-injection and SSRF defenses. Applied guardrails in order to ensure quality responses.
- Built a multilingual **storefront AI salesman**: 8 tools, vector product search over Qdrant, **13 languages including 10 Arabic dialects**, streaming responses. **60K+ real conversations**.
- Build a shared **nanoUSD credit ledger** with settle - deploy - void patterns, that auto-refunds on failures and ensure MongoDB **atomicity**.
- Built the **Landing Page Generator**, a multi-agent image pipeline turning product photos into finished creatives. **1,000+ generations** in its first month as a paid feature.
- Built a seller-facing, spec-compliant **MCP server** exposing **57 tools** with oauth scopes and strict single-tenant isolation.
- Built a Go-powered **real-time store analytics** on **ClickHouse** ingesting **4M+ events/hour** across all stores, with **RabbitMQ** ingestion and a 7-view materialized aggregation pipeline.

Amaris Consulting - PFE Internship Feb 2025 – Aug 2025
AI-Enhanced Modular Recruitment Platform
Technologies: GenAI, NLP, LLMs, Python, Microservices, Docker

- Built an AI-powered microservice-based recruitment system using **LLMs and NLP** with transparent scoring, and automated **CV parsing** from LinkedIn and PDF sources. Orchestrated through **Jenkins**, and deployed on Docker Swarm.

EY - Tutored Project Sep 2024 – Dec 2024
Optimization of Data Structures Testing Using Generative AI
Technologies: LangChain, ML, NLP, Python

- Automated data-structure testing with **open-source agents** for schema optimization and validation. Utilized **chunking** techniques, and prompt engineering to ensure quality results.

Projects

Scalable Slack-Clone with Docker Swarm & Jenkins CI/CD Jan 2024 – May 2024

- Containerized a real-time chat app with **Docker Swarm** auto-scaling, a **Jenkins CI/CD** pipeline, and Prometheus/Grafana monitoring.

Skills

Generative AI LLMs, RAG, LangChain, Vercel AI SDK, MCP	DevOps Kubernetes, Docker, GitHub Actions, OpenResty
Models & Vector Vertex AI, Gemini, OpenAI, Qdrant	Languages Python, TypeScript, JavaScript, Java, C++ , Go
Backend TypeScript, Node.js, Express, MongoDB, Redis	Cloud AWS, Azure, GCP
ML & Data TensorFlow, Prophet, SageMaker, Kafka, Spark	Spoken Arabic, French, English (TOEIC 990/990)

Certifications

NVIDIA: Building RAG Agents with LLMs Nov 2024

AWS Academy: Machine Learning Foundations, Data Engineering Oct 2024

freeCodeCamp: Data Analysis, Data Visualization, ML with Python 2022 – 2023